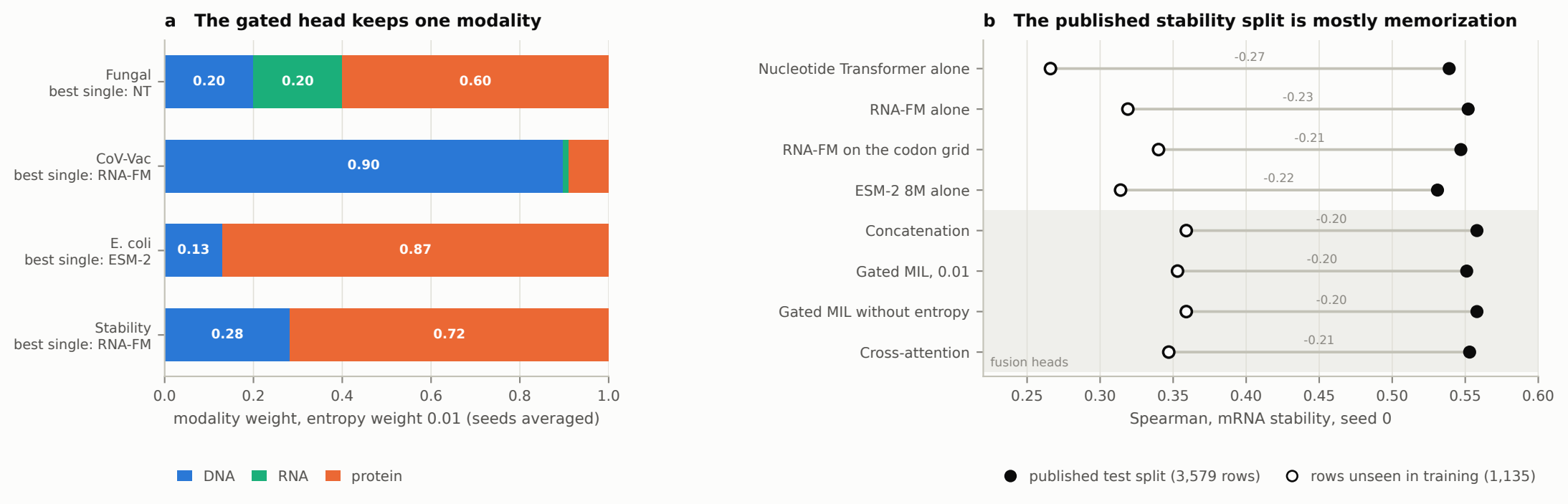
a The gated head keeps one modality
Fungal best single: NT
CoV-Vac best single: RNA-FM
E. coli best single: ESM-2
Stability best single: RNA-FM
0.20
0.20
0.60
0.90
0.13
0.87
0.28
0.72
0.0
0.2
0.4
0.6
0.8
1.0
modality weight, entropy weight 0.01 (seeds averaged)
DNA
RNA
protein
b The published stability split is mostly memorization
Nucleotide Transformer alone
RNA-FM alone
RNA-FM on the codon grid
ESM-2 8M alone
Concatenation
Gated MIL, 0.01
Gated MIL without entropy
Cross-attention
fusion heads
-0.27
-0.23
-0.21
-0.22
-0.20
-0.20
-0.20
-0.21
0.25
0.30
0.35
0.40
0.45
0.50
0.55
0.60
Spearman, mRNA stability, seed 0
published test split (3,579 rows)
rows unseen in training (1,135)